

A solid red circle.

FS D7000 Series Raman Amplifier Data Sheet

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| Product Overview

The D7000 Raman Amplifier is high-power, low-noise designed for distributed Raman amplification, offering cost-effective solutions to extend the optical link power budget and significantly enhance OSNR. By using transmission fiber as the gain medium, it achieves distributed amplification, reducing system noise while optimizing gain and noise performance. Supporting from a single DWDM channel to up to 80/96 channels, the D7000 Raman Amplifier meets the demanding requirements of service providers and enterprise networks, ensuring superior reach and optical performance.

The D7000 series is a 200G/400G stackable wavelength division transmission platform featuring large transmission capacity, small size, low power consumption, and easy operation and maintenance. It provides highly scalable, reliable, and secure connectivity, and can be widely used in data center interconnections and new metropolitan area network transmissions. The D7000 supports AmpCon™-T/WebGUI/CLI management, which enables service configuration and performance monitoring, alerts and optical power monitoring in real time to help you manage your network and devices more simply and conveniently.

| Features and Benefits

- Up to 6-14dB Average Gain for G.652 Fiber
- Distributed Raman Amplification with Low Noise Figure
- Dual AC/DC Pluggable Power Supply and Pluggable Fan Unit
- Support WebGUI, AmpCon-T Management
- Support Up to 80/96 DWDM Wavelengths
- Pump Polarization-independent Design
- Suitable for Long-span or High-speed Optical Transmission

Product Details

Product configurations



Figure 1.

D7000 BDRA5008 Co-propagating Distributed Raman Amplifier with 8dB Gain, 80 Wavelengths, Dual AC PSUs, 1U Rack Mount



Figure 2.

D7000 PDRA5014 Counter-propagating Distributed Raman Amplifier with 14dB Gain, 80 Wavelengths, Dual AC PSUs, 1U Rack Mount

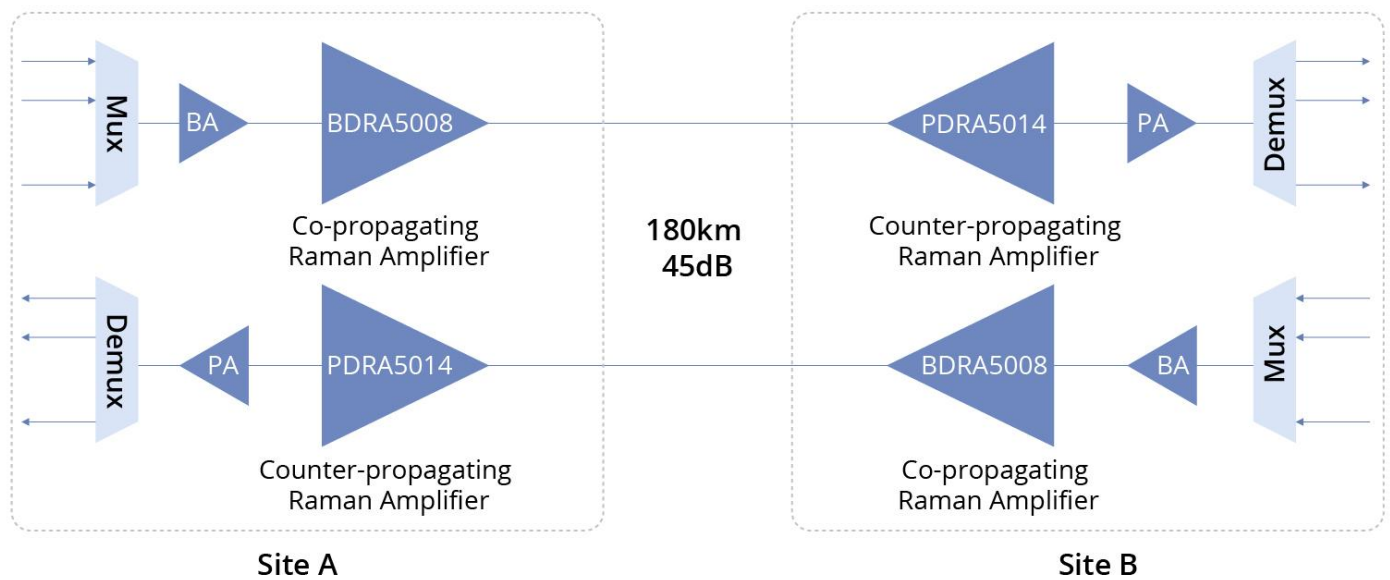


Figure 3.

Customized Distributed Raman Amplifier, Dual AC/DC PSUs, 1U Rack Mount

Application

D7000 raman amplifiers are widely used in long-haul and ultra-long-haul fiber optic transmission systems, high-capacity DWDM systems, and high-speed optical communications. They provide distributed gain to compensate for signal loss, extend transmission distances, and support multiple wavelength channels, enhancing system capacity and signal quality. Additionally, raman amplifiers perform well in remote or harsh environments and are compatible with various fiber types, ensuring the stability and reliability of optical networks.



Technical Specification

Parameter	BDRA5008	PDRA5014
Optical Specifications		
Product Type	Co-propagating	Counter-propagating
Operating Wavelength @80CH	1529~1561nm (1510nm typical)	1529~1561nm (1510nm typical)
Signal Input Power*	0~20dBm	-45~-20dBm
Pump Wavelength	1425~1490nm (1480nm typical)	1425~1490nm (1480nm typical)
Pump Output Power	1000mW	1000mW
ON/OFF Gain*	8dB	14dB
Gain Flatness	<2.5dB	<2.5dB
Polarization Dependent Gain	0.3dB	0.3dB
Polarization Mode Dispersion	0.3ps	0.3ps
Noise Figure	-3~0dB (-2dB typical)	-3~0dB (-2dB typical)
OSC Channel	Support 1510nm/1625nm (Supported); 1490nm (Not supported)	Support 1510nm/1625nm (Supported); 1490nm (Not supported)
Operation Mode	ACC	ACC
Port		
Connector	LC/APC	LC/APC
Management Interface*	1x ETH RJ45 1x RS485 1x RS-232	1x ETH RJ45 1x RS485 1x RS-232
Management		
Management Type	Device buttons, AmpCon-T, WebGUI	Device buttons, AmpCon-T, WebGUI

Note:

1. If you need a counter-propagating raman amplifier with power lower than -45dBm, please [contact your sales](#) to confirm the relevant technical specification parameters.
2. You can adjust the ON/OFF gain through the network management, the adjustment range can be from 0~8dB (co-propagating raman), 0~14dB (counter-propagating raman), but after the adjustment, it will lead to a change in the gain flatness of the whole device.
3. RS485 and RS-232 are debug ports, not open to the public.

Technical Specification

Parameter	BDRA5008	PDRA5014
Physical Specification		
Power Supply	85VAC~123VAC;170VAC~264VAC	85VAC~123VAC;170VAC~264VAC
Power Consumption	30W	30W
Weight	6kg	6kg
Dimensions (HxWxD)	44x236x483mm	44x236x483mm
Environmental		
Operating Temperature	-5 to 60° C (23 to 140° F)	-5 to 60° C (23 to 140° F)
Storage Temperature	-40 to 80°C (-40 to 176° F)	-40 to 80°C (-40 to 176° F)
Relative Humidity	5%~85% (Non-condensing)	5%~85% (Non-condensing)
MTBF	≥ 20 years (at 25°C operating temperature)	

Technical Specification

Parameter	DRA5000
Optical Specifications	
Product Type	Co-propagating/Counter-propagating
Operating Wavelength	1529~1561nm (80CH) 1528~1567nm (96CH)
Signal Input Power*	0~20dBm (Co-propagating) -45~-20dBm (Counter-propagating)
Pump Wavelength	1425~1490nm (1480nm typical)
Pump Output Power	1000mW
Custom Gain Range*	6~14dB
Gain Flatness	<2.5dB
Polarization Dependent Gain	0.3dB
Polarization Mode Dispersion	0.3ps
Noise Figure	-3~0dB (-2dB typical)
OSC Channel	Support 1510nm/1625nm (Supported); 1490nm (Not supported)
Operation Mode	ACC
Port	
Connector	LC/APC;LC/UPC;SC/APC
Management Interface*	1x ETH RJ45 1x RS485 1x RS-232
Management	
Management Type	Device buttons, AmpCon-T, WebGUI

Note:

1. If you need a counter-propagating raman amplifier with power lower than -45dBm, please [contact your sales](#) to confirm the relevant technical specification parameters.
2. You can customize the ON/OFF gain in 6dB, 8dB for co-propagating raman, 10dB, 12dB, 14dB for counter-propagating raman, for other range of switching gain values, please [contact sales](#) for customization.
3. RS485 and RS-232 are debug ports, not open to the public.

Technical Specification

Parameter	DRA5000
Physical Specification	
Power Supply	AC: 85VAC~123VAC;170VAC~264VAC DC: -40V~-60VDC
Power Consumption	30W
Weight	6kg
Dimensions (HxWxD)	44x236x483mm
Environmental	
Operating Temperature	-5 to 60° C (23 to 140° F)
Storage Temperature	-40 to 80°C (-40 to 176° F)
Relative Humidity	5%~85% (Non~condensing)
MTBF	≥ 20 years (at 25°C operating temperature)

Ordering Information

FS P/N	Product Description
Chassis	
D7000-CH2U	D7000-CH2U, 8 Slots 2U Stackable Unloaded Chassis, Support Single-wavelength 200G/400G, Up to 3.2Tbps Capacity, Redundant AC PSUs, Support SDN Network Management
D7000-CH2U	D7000-CH2U, 8 Slots 2U Stackable Unloaded Chassis, Support Single-wavelength 200G/400G, Up to 3.2Tbps Capacity, Redundant DC PSUs, Support SDN Network Management
Muxponder	
4MC4	D7000 4MC4, 1 x 400G QSFP-DD or 4 x 100G QSFP28 to 1 x 400G CFP2 Muxponder for D7000 Managed Chassis
2MC2	D7000 2MC2, 2 x 100G QSFP28 to 1x 200G CFP2 Muxponder for D7000 Managed Chassis
11MC2	D7000 11MC2, 1 x 100G QSFP28+10x 10G SFP+ to 1x 200G CFP2 Muxponder for D7000 Managed Chassis
20MC2	D7000 20MC2, 20 x 10G SFP+ to 1 x 200G CFP2 Muxponder for D7000 Managed Chassis
EDFA	
OA1825	D7000 OA1825, DWDM EDFA Bidirectional Amplifier with PA 15~25dB and BA 8~18dB Gain , 21.5dBm Output for D7000 Series Managed Chassis, with Built in OSC, VOA
OA1835	D7000 OA1835, DWDM EDFA Bidirectional Amplifier with PA 22~35dB and BA 8~18dB Gain , 21.5dBm Output for D7000 Series Managed Chassis, with Built in OSC, VOA
OLA2525	D7000 OLA2525, DWDM EDFA In-line Amplifier with Dual LA 15~25dB Gain, 21.5dBm Output for D7000 Series Managed Chassis, with Built in OSC, VOA
Raman Amplifier	
PDRA5014	D7000 PDRA5014 Counter-propagating Distributed Raman Amplifier with 14dB Gain, 80 Wavelengths, Dual AC PSUs, 1U Rack Mount
BDRA5008	D7000 BDRA5008 Co-propagating Distributed Raman Amplifier with 8dB Gain, 80 Wavelengths, Dual AC PSUs, 1U Rack Mount
DRA5000	Customized Distributed Raman Amplifier, Dual AC/DC PSUs, 1U Rack Mount
MUX&DEMUX	
OMD48ECM	OMD48ECM, 48 Channels 100GHz C14-C61 Passive, Dual Fiber DWDM Mux and Demux with Monitor Port, LC/UPC, 1U Rack

Ordering Information

FS P/N	Product Description
OLP	
OPB2-I	D7000 OPB2-I, 1+1 Optical Protection Switch for D7000 Managed Chassis, with Built in 2 Protection Groups
OPB2	D7000 OPB2, 1+1 Optical Protection Switch for D7000 Managed Chassis
OTDR	
OTDR08	D7000, OTDR08, 8 Channels 1501nm Optical Time Domain Reflectometer (OTDR) for D7000 Series Managed Chassis
OCM	
OCM08	D7000 OCM08, 8 Channels 1528~1568nm Optical Channel Monitor (OCM) for D7000 Series Managed Chassis
ROADM & OADM	
ROADM-09T	D7000 ROADM-09T, 9-degree CDCF Pluggable ROADM for D7000 Managed Chassis, Featuring WSS Technology
TFF04	Customized 4CH Dual Fiber DWDM OADM for D7000 Series Managed Chassis
Accessories	
D7000-PSU-AC	Hot-swappable AC Power Module 1300W for D7000 Series Managed Chassis
D7000-PSU-DC	Hot-swappable DC Power Module 1000W for D7000 Series Managed Chassis
D7000-FAN	Hot-swappable Fan Module, Front-to-Back Airflow through the D7000 Series Managed Chassis
D7000-AUX	D7000-AUX, Interface Module for D7000 Series Managed Chassis, 4 x RJ45 Ports
D7000-EMU	D7000-EMU, Console for D7000 Series Managed Chassis
D7000-FILLER	D7000-FILLER, Port Filler for D7000 Series Managed Chassis
CFP2-DCO-400G-D	400G DWDM Tunable Coherent CFP2 DCO 80km DOM Duplex LC SMF Transceiver Module For Transmission, Used with D7000 Series
CFP2-DCO-200G-D	200G DWDM Tunable Coherent CFP2 DCO 80km DOM Duplex LC SMF Transceiver Module For Transmission, Used with D7000 Series
AmpCon-T Network Management	
LIS-AMPCON-T-F	AmpCon-T Management Platform for FS OTN Devices, Includes Visual Monitoring, Centralized Management and Intelligent Analysis

I Additional Information

For more information about the D7000 Series, contact your account manager or visit

https://www.fs.com/search_result?keyword=D7000

I Document History

New or revised topic	Described in	Date
Updates to FS D7000 Raman Amplifier Data Sheet	Updated all	9/10/2024



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